AD 502 – Machine Learning

Pre-Reading Material – Lecture 33

Random Forest-I

Introduction to Random Forest

What Is Random Forest?

Random Forest is a supervised machine learning algorithm used for classification and regression tasks. It is an **ensemble learning technique** that combines multiple decision trees to produce a final prediction.

A single decision tree may overfit the training data or produce unstable predictions. Random Forest reduces this limitation by combining the predictions of several trees.

What Is Ensemble Learning?

Ensemble learning is a machine learning approach in which multiple models work together to solve a problem. The combined prediction is often more reliable than the prediction of an individual model.

Random Forest is an example of ensemble learning because it builds multiple decision trees and combines their outputs.

What Is a Decision Tree?

A decision tree is a supervised learning algorithm that makes predictions through a sequence of decision rules. It consists of a root node, decision nodes, branches and leaf nodes.

For example, a decision tree may classify a student as likely to pass or fail based on attendance and previous examination marks.

Why Is Random Forest Needed?

An individual decision tree can be sensitive to changes in training data. Small changes in the data may produce a different tree and different predictions.

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Random Forest addresses this problem by constructing multiple trees using different samples and subsets of features. Combining their predictions generally improves stability and can reduce overfitting.

Applications of Random Forest

Random Forest is used in:

- Student performance prediction.
- Disease classification.
- Fraud detection.
- Weather and electricity-demand forecasting.
- House-price prediction.

Conclusion: Random Forest combines multiple decision trees to produce reliable predictions for different machine learning problems.

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